



Chatbots, search engines, and the sealing of knowledges

Nora Freya Lindemann¹ 

Received: 16 January 2024 / Accepted: 3 April 2024 / Published online: 22 April 2024
© The Author(s) 2024

Abstract

In 2023, online search engine provider Microsoft integrated a language model that provides direct answers to search queries into its search engine Bing. Shortly afterwards, Google also introduced a similar feature to its search engine with the launch of Google Gemini. This introduction of direct answers to search queries signals an important and significant change in online search. This article explores the implications of this new search paradigm. Drawing on Donna Haraway's theory of *Situated Knowledges* and Rainer Mühlhoff's concept of *Sealed Surfaces*, I introduce the term *Sealed Knowledges* to draw attention to the increasingly difficult access to the plurality of potential answers to search queries through the output of a singular, authoritative, and plausible text paragraph. I argue that the integration of language models for the provision of direct answers into search engines is based on a de-situated and disembodied understanding of knowledge and affects the subjectivities of its users. At the same time, the *sealing of knowledges* can lead to an increasing spread of misinformation and may make marginalized knowledge increasingly difficult to find. The paper concludes with an outlook on how to resist the increasing *sealing of knowledges* in online search.

Keywords Chatbots · Large language models · Search · Situated knowledge · Sealed surfaces · Sealing of knowledges

1 Introduction

In autumn 2022, a newly released AI system caused a massive public debate and hype: ChatGPT. Developed by the company OpenAI, ChatGPT is a chatbot which, in its basic version, can be used for free.¹ It is an open domain chatbot with which users can 'chat' about a variety of topics, ranging from quantum physics to science fiction novels to cake recipes. Immediately after its release, a debate broke out about the potential uses of this technology and the impact it could have on various aspects of life, such as education, various professions, and online search. Microsoft soon announced an agreement with OpenAI to integrate ChatGPT into its online search engine "Bing" (Lindern 2023; Milmo 2023). It is not the only company sensing financial potential in the integration of a chatbot into its search engine. The Chinese search engine giant Baidu, too, announced that it intends to integrate its chatbot "ERNIE" into its search engine (Gusbeth 2023; Huang 2023). Moreover, Google, the company

providing the globally most widely used search engine, also announced its plans to integrate a chatbot into its search engine and did so in late 2023 through the launch of Google Gemini (Google; Pichai and Hassabis, December 6th, 2023). With this current trend, it is crucial to take a step back and question the implications of the deployment of chatbots to generate direct responses to online searches. While there has been a critical discourse on this topic before (as for example the illuminating paper "Situating Search" by Shah and Bender (2022) proves), philosophical and ethical discussions of it remain sparse.

This research gap on one hand, and a growing number of people using chatbots providing direct answers to search queries on the other, are the starting point of this paper, in which I discuss the impact of this use of chatbots on users and the findability of knowledges. Working with the theory of *Situated Knowledges* by Donna Haraway and the concept of *Sealed Surfaces* by Rainer Mühlhoff, I argue that utilizing chatbots to provide direct answers to search queries can lead to an increasing *Sealing of Knowledges*. The term *Sealed*

✉ Nora Freya Lindemann
norafreya.lindemann@uni-osnabrueck.de

¹ Institute of Cognitive Science, University of Osnabrück, Osnabrück, Germany

¹ There is also a more advanced version which users can access by paying for a subscription. In the basic version, users still have to log in via an account with OpenAI or an existing account with e.g. Microsoft. This allows OpenAI to collect data about its users, which could put the concept of free access up for debate.

Knowledges draws attention to the increasingly difficult access to the multitude of possible answers to search queries through the output of a singular, authoritative, and plausible text paragraph by a chatbot integrated into an online search engine. The *sealing of knowledges* creates specific user subjectivities and may particularly impact the findability of marginalized knowledges which leads to the reproduction of hegemonic knowledges. Moreover, the usage of chatbots as search engines is based on a specific, de-situated and disembodied understanding of knowledge. By introducing the concept of *sealed knowledges* and by problematising the understanding of knowledges on which the implementation of chatbots into search engines is based, I introduce a new perspective and analytical framework to the critical debate on this mode of online search.

The paper is structured as follows. It starts with a discussion and explanation of its central terms: Chatbots, *situated knowledges* and *sealed surfaces*. I then introduce the concept of *sealed knowledges* to describe the effect of integrating direct answers in online search on the plurality of the possible answer space to a search query. Followingly, both the underlying assumptions about users of search engines and the effect of direct answers on their subjectivities are discussed. Afterwards, the assumption of knowledges and the findability of different knowledge in this mode of search are analysed. The paper concludes with an outlook on how to resist the increasing *sealing of knowledges*: It is important to remain doubtful, to re-situate the understanding of knowledge and to create an awareness for the increasing *sealing of knowledges*.

2 Chatbot as search engines: sealing knowledges

2.1 Search engines, large language models and chatbots

AI algorithms based on machine learning (ML) have already been used in search engines for years. They determine for example in which order search results are arranged and which advertisements are shown to which users in which order. A newer development is the integration of AI algorithms for language processing, so called language models (LMs) or large language models (LLMs), which are trained on large quantities of text and consist of a neural network with a very large number of parameters, into search engines. They have been employed to process user inquiries and to extract which information a user is looking for (Nayak 2019).² The recent development of integrating LLMs

for language generation, i.e. for producing direct textual responses to online search queries goes an important step further as it leads to LMs being used not only for *query understanding* but also for *answer generation*. This development, of LLMs being used to auto-generate textual answers to user input in search engines, is the current focus of search engine development and the focus of this article. To indicate that this paper specifically refers to the automated generation of textual answers based on user input—and not only the sense-making of user input—the term ‘chatbot’ is used.

2.2 Situated, partial and embodied knowledges

Another term is central to this paper: *knowledges*. By using the term *knowledges* (in plural), I situate my research in a specific theoretical tradition. Since the beginning of the 1980s, feminist theorists, philosophers, writers and activist increasingly turned to questions of *what* constitutes knowledges, *how* knowledges are produced, and *which and whose* knowledges are considered to be knowledge (Anderson 2020). These questions have a political dimension as they are intertwined with the question who has—and who has not—the power to define common paradigms and knowledges in a society. Feminist epistemologists and theorists, with this political understanding of knowledge, divert from dominant conception of knowledge as impartial, neutral and detached from its context of ‘discovery’ (Harding 1986; McLaren 2002; Stanley and Wise 2013; Traweek 1988; Waldbry 1995). More than just diverting from it, they pose a critique on visions of knowledge as ‘objective’, which posit that knowledge is independent from the person formulating knowledge claims and therefore disembodied (Anderson 2020). This, critical epistemologists claim, lacks an awareness of the inherent political nature of knowledge, as the question of which and whose knowledge is considered knowledge is always also a question of power. It matters who produces knowledges and whose knowledge is regarded as knowledges.

Donna Haraway (1988) prominently wrote about knowledges and possibilities of knowing. In her seminal paper *Situated Knowledges: The Science Question in Feminism and the Privilege of Partial Perspective* from 1988, Haraway formulated her theory of *situated knowledges*. The theory holds that knowledges are always situated and partial. There are no knowledges that are detached from the person who formulates them, and no knowledges that are independent of the respective cultural, historical, and social

Footnote 2 (continued)

traveler to usa need a visa” (Nayak 2019). In this search, the “to” is decisive as it indicates that the user is looking for information on visa *from Brazil to the USA* and not the other way around.

² The language models are used to decipher what users are looking for in longer search queries by taking prepositions like “to” and “for” into account. This matters for example in the search query “brazil

context in which they are produced. Turning against dominant narratives and understandings of objective knowledge as independent from the person making knowledge claims, Haraway (1988, p. 583) positions herself and her theory in opposition to what she terms the *god trick*. The god trick is the claim to see “everything from nowhere” (Haraway 1988, p. 581), the already mentioned dominant idea that there are impartial, de-situated, disembodied, whole truths and knowledges. Instead, Haraway argues that knowledges are always produced by humans, may it be through scientific experiments, through try and error, through thought, or through experience. We see, hear and experience with our bodies. Bodies shape which experiences we make and thus what we know. Only with our bodies can we come to formulate knowledges. We are perceived by others through our bodies and the knowledge that is accepted as knowledge within society often depends on which body a person has. A person who is Black will be perceived differently when making knowledge claims than a white person (and is less likely to be believed), just as there are differences in credibility assigned to people of different genders and different bodily abilities. At the same time, people who are racialized and gendered make different experiences within society than people who pass as unmarked, leading to them carrying different knowledges. With that, knowledges are always essentially partial, as humans have different experiences, different pre-knowledges, different bodies. Feminist objectivity, then, in Haraway’s understanding, “is about limited location and situated knowledge, not about transcendence and splitting of subject and object”.

With this, she moves away both from ideas of knowledge as independent of the person who carries it and from previous ideas of subjugated knowledges which Harding (1986, p. 28) proposed. Theories of subjugated knowledge claim that it is especially the knowledges of subjugated and marginalized people that need to be appreciated and listened to. While Haraway agrees that subjugated knowledges should be acknowledged and listened to, she does not consider them as inherently more important than others. Rather, it is crucial is to appreciate and acknowledge the value of partial perspectives which are grounded in their own situatedness. This necessarily includes an appreciation of a “view from a body, always a complex, contradictory, structuring, and structured body, versus the view from above, from nowhere, from simplicity” (Haraway 1988, p. 589).

While this may at first sound like there are only individual knowledges, this is not what Haraway has in mind. *Situated Knowledges*, on the very contrary, allow for “the joining of partial views and halting voices into a collective subject position that promises a vision of the means of ongoing finite embodiment, of living within limits and contradictions—of views from somewhere” (Haraway 1988, p. 590). Knowledges are situated and there is a plurality of different

knowledges. However, importantly, they can and do join into knowledge collectives. This allows for a solidarity and collective knowledges while avoiding the *god trick*.

I use the term *knowledges* in the plural to show the always already existing plurality of (possible) knowledges which are produced by embodied subjects, always partial, always situated, yet joining and forming coalitions. They develop in certain conditions and structure realities in powerful ways. *Knowledges* are something a person knows. It entails an understanding of a phenomenon and the ability to apply an information to something else or to act on it. It has always an embodied character as a person’s knowledge is tied to their embodied experience. *Information*, in contrast, is a something that is e.g. written down somewhere and can be accessed by a person to gain knowledge. It is broken down so that it can be transferred and processed. With that, it is lacking context and application. Importantly, information in that sense is not de-situated either as it has been written down as a knowledge someone previously gained. I consider knowledges and information as always situated. The predominant usage of the term *knowledges* in this paper signifies its theoretical rooting in Haraway’s concept of *Situated Knowledges*.

2.3 Sealed surfaces

The usage of chatbots providing direct answers to search queries profoundly changes how and which knowledges users can find in internet searches. It leads to what I call *sealed knowledges*. The terminology and metaphor of *sealing* derives from the concept of *sealed surfaces*, which Rainer Mùhlhoff (2018) introduced to refer to.

the observation that the design of digital devices often conceals the technical nature of the device beneath closed surfaces, which manifest as user interfaces. The design of human-machine interfaces is increasingly involving the use of design features that hinder users from gaining insights into the technical workings of devices and applications. (Mùhlhoff Forthcoming, p. 50)

The term is both a description and critique on the current trend in UX (user experience) design which conceals the inner functioning of (digital) devices. This trend has two aspects: First, “users are not supposed to—or assumed not to want to—use a (digital) device instrumentally, meaning as a tool with a generally indefinite list of purposes, which the user can operate through conscious reflection and knowledge of the device’s technical characteristics” (Mùhlhoff Forthcoming, p. 55). In UX design, users are not expected to use electronic devices as purpose open tools whose usability can be understood through knowledges about their technical characteristics. This goes hand in hand with a very specific image of a user who acts on the stimuli presented by the

intuitive interfaces and does not aim to act on it because of a comprehension of its function. This is related to the second aspect of the sealed surfaces, namely that it is not considered reasonable that users have the will, desire, or ability to structurally understand the functions of a device (Mühlhoff 2018, p. 568).

For example, Microsoft has changed the Windows folder structure over the years in such a way that the inner workings of the system became increasingly obscure in newer versions. The critique of *sealed surfaces* contents that in UX design trends of the last decades, the rationale seems to be that non-experts do not need to access deeper folder levels and thus to understand how the system works (Mühlhoff 2018, p. 566f.). As a result, users must put more and more effort into familiarizing themselves with how the system works to use it instrumentally, while the interfaces are increasingly designed in such a way that it is unattractive for users to do so. While knowledgeable users can still get through the “sealed surfaces” and e.g. understand how the devices whose surface is sealed functions internally, it is less and less common knowledge of average users. This growing inaccessibility is the starting point of Mühlhoff’s critique on UX design and his concept of *sealed surfaces*. While the surfaces are *sealed*, thus difficult to get through, the seal can be broken, and the surfaces can be permeated by knowledgeable users who are willing and able to put effort into it. This is why the metaphor of *sealing* is so apt. Mühlhoff (2018) argues that this trend of *sealed surfaces* is part of UX design and its call for usability, effectivity, efficiency and user satisfaction without the aim that users understand the systems. As I will show in this paper, the metaphor of *sealing* is also helpful to understand the impact that chatbots used to assemble direct answers in search engines have on knowledges.

3 Sealed knowledges

While Mühlhoff (2018) refers to (digital) device interfaces with the term *sealed surfaces*, the term *sealed knowledges* points towards the increasing *sealing of knowledges* through the usage of chatbots providing direct answers to search queries. Before turning to a more detailed explanation of the term *sealed knowledges*, I will briefly situate the current trend in online search paradigms historically. This sheds light on the (dis)continuities of online search, what has changed through the introduction of chatbots providing direct answers, and why the concept of *sealed knowledges* is important to describe this current trend. Online search started to be widely used in the 1990s with the growing personal use of computers and the internet by the public. The number of websites that could be explored was a lot smaller back then in comparison to today. Results of online search queries were randomly displayed in search engines. There

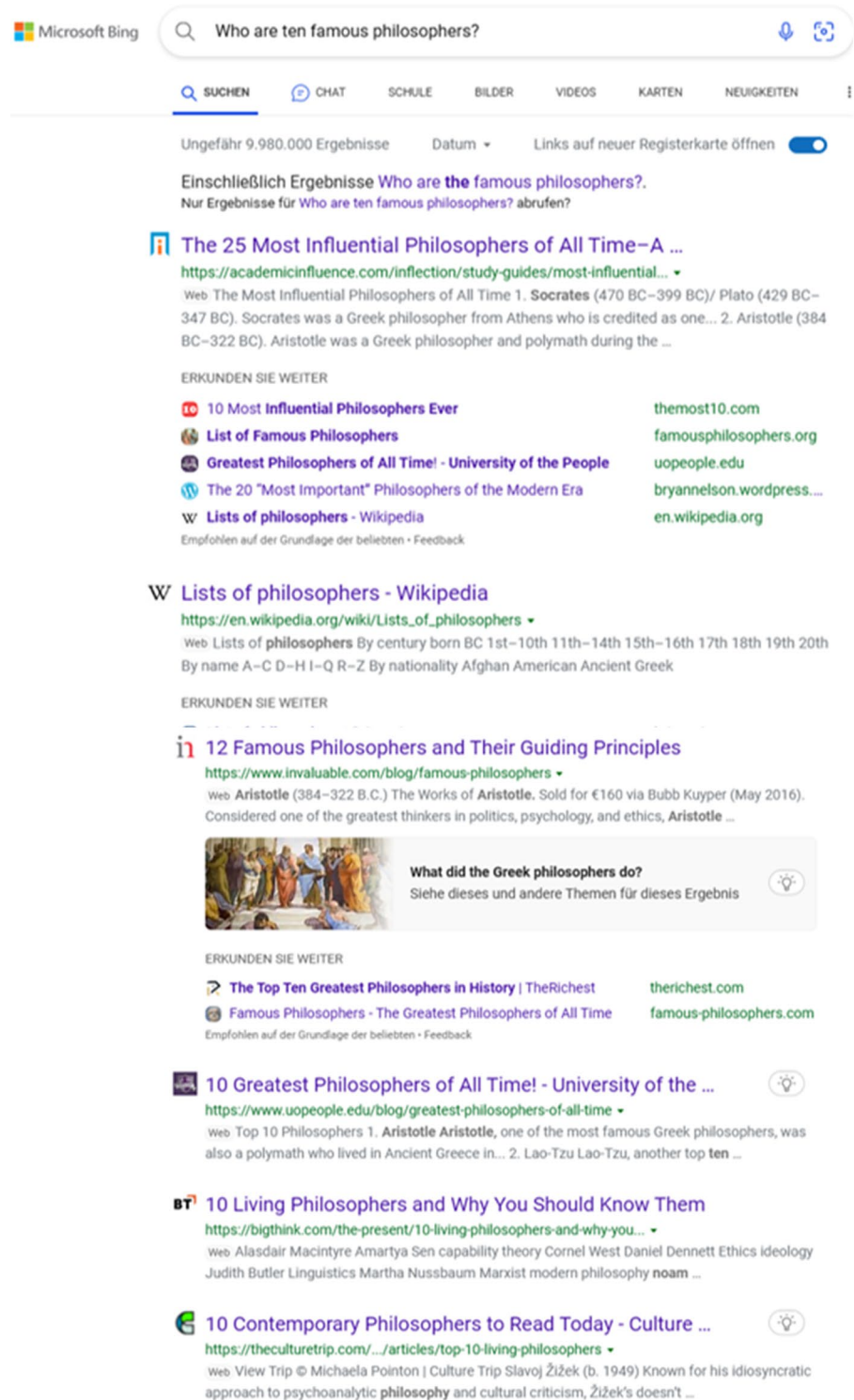
was no specific order in which they were shown to users. This changed already in 2000, when the search results were first structured according to which information is (or rather: could be) most useful for users (Introna and Nissenbaum 2000). The introduction of an algorithm structuring the display of search results has an important impact on which information is found the most. Over 25 percent of Google users for example click on the first displayed search result (Southern 2020) and only under 5 percent of users go to the second page of search results (Reputation911 2023).

The algorithms structuring search thus carry significance and politics as they crucially mitigate which web pages are shown in which order and with that which knowledges are dominantly found (Introna and Nissenbaum 2000; Segev 2008; van Dijck 2010). This means that the display of search results in what I will call ‘traditional’ web searches, in which no direct answers are assembled by a chatbot, is (1) mediated according to algorithms of the search engines companies which are structuring them not the least according to financial incentive, is (2) very important to what knowledges are dominantly found, and therefore, (3) carries political implications and has a powerful position in the societal knowledge infrastructure. The answers the search engine outputs impacts on societal knowledges and ways of knowing (Röhle 2010; Segev 2008). When looking for information, most people turn to online searches to find them. The traditional role of books and libraries declined over the last decades while online searches took over their role. They provide a way faster and, arguably, easier accessible way of finding knowledges.

This is similar in the usage of chatbots providing direct answers to search queries, which are mostly integrated in already existing search engines and therefore belong to private, large cooperations. However, the provision of direct answers in online search has qualitative differences to ‘traditional’ search, which I summarize under the term *sealed knowledges*. Through the usage of Chatbots providing direct answers, the complexity of the possible answer space, the plurality of potential answers to a search query, is increasingly sealed. Consider Figures 1 and 2 which show the answers for the search query “Who are ten famous philosophers?” both in the ‘regular’ Bing search function (Fig. 1) and the Bing chat function providing direct answers (Fig. 2).³ In Fig. 1, the ‘traditional’ web search displays ‘pages’ of search results, different links to websites that may contain useful and helpful information to answer the search query. In this specific case, once clicking on the different links, different results

³ I took the inspiration for this example from a Twitter (now X) post by Katie Stockdale who inputted a similar question into ChatGPT (Stockdale, February 27th 2023).

Fig. 1 Screenshot, June 2023. Bing's 'traditional' search interface when the question "Who are ten famous philosophers?" is entered



can be obtained. The first entry does not provide a list of 10 famous philosophers, but directly shows a list of 25 famous philosophers. The second entry, Wikipedia, portrays a page on which links lead to different sub-pages of famous philosophers which are sorted according to various aspects such as geographical location of the philosophers,

their school of thought, their gender, whether they are still alive, etc. Many of the initial links in the web search provide lists of more than ten philosophers. The headings of the different pages already give a hint of what the users will find once they click on them (e.g. a list of living philosophers).

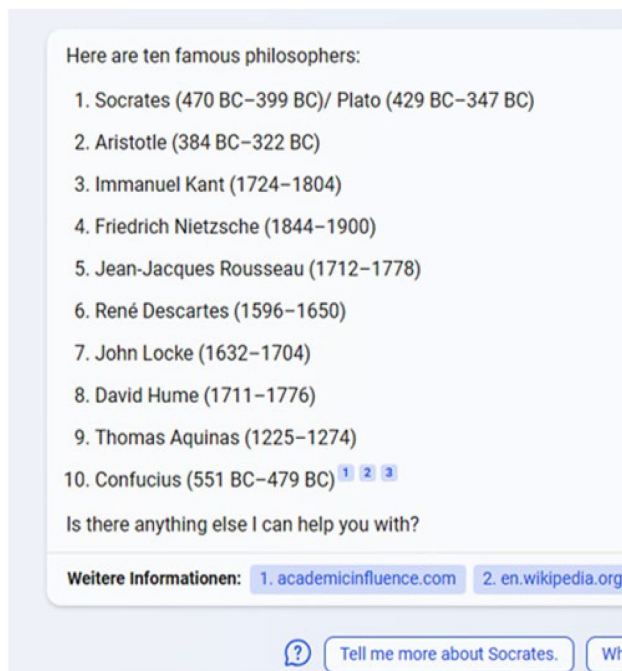


Fig. 2 Screenshot, June 2023. Bing's chat interface when the question "Who are ten famous philosophers?" is entered

The Bing chat function in Fig. 2 in contrast outputs an easily understandable, concise list of names of people which are deemed famous philosophers. Their names are in a ranked list without any further information. In this example, there is not a single living philosopher, female philosopher, philosopher from Africa, or the Americas named. The sheer potential that there may, or could, be different answers to the search query is not visible or perceptible in the seemingly authoritative and conclusive answer of the chatbot. Below the answer are links to websites on which users can find some further information about the search query (see also Metzler et al. 2021). However, while it is possible to click on them for information verification, it seems rather unlikely that users will regularly do so. Users decide to use the chat function of Bing precisely to receive a singular answer without having to look at several different sources. Moreover, the answer of the bot seems conclusive. Why putting the time and effort into looking for further knowledges and for information verification? In addition, following the answer to the initial search there are directly proposed follow-up searches that users can decide to click on (see Fig. 2). Socrates is the first on the list of famous philosophers—and users interested in him, and his philosophy, can click on the conveniently pre-formulated search query "tell me more about Socrates" without even having to put effort into typing the search query themselves. The initial response already received in the first search query can be expanded in this way. The

follow-up search proposed by the search engine is thus closely related to the first search result.

In the direct provision of a search result, the plurality of the possible answer space and of possible knowledges that could be relevant for a search, are *sealed*. Different than in 'traditional' web search, the output of a single answer does not show the contingency of the knowledges it conveys. Answers to search queries are often complex and there is not only one, true answer to a search query. By showing only one answer, the computer scientists Potthast et al. (2020, p. 8) argue, "a strong message is sent to the user: 'There's nothing else to be said'. Obviously, much else needs to be said in answer to these particular questions, but the format of direct answers is entirely insufficient to do so". The ten philosophers the Bing chat function outputs in Fig. 2 are mostly agreed upon famous philosophers. But there are many more famous philosophers. Their names, however, are not shown in the search result, and even the potential to look for them is not clear as there are no website headings pointing in their direction. In addition, the proposed further searches are building on the already given answer and thus do not invite to expand on the first search result, for example by asking for famous living philosophers.

We live in a complex world and answers to questions are mostly context dependent. Therefore, there is often not one true, right, and good answer to a question. Quite on the contrary, the answer to complex question is mostly rather "it depends" (Potthast et al. 2020). As Potthast et al. (2020, p. 10) argue in connection to AI produced answers to search queries: "For many types of questions and retrieval tasks no direct answers should be given due the complexity of their underlying hypothesis space." This complexity and context dependency of possible answers to search queries are not mirrored in chatbots providing direct answers. The chatbot algorithm necessarily assembles pre-existing texts in new ways, take it out of context and summarize the knowledges, breaking the answer space down tremendously. Therefore, the complexity and ambiguity of the answer space and the process of the automatic answer generation is *sealed* which may easily lead to the wrong impression that the chatbots gives the one, true answer. Moreover, through the usage of chatbots providing direct answers to search, what Shah and Bender (2022) term the "serendipity of search", the potential for stumbling across related, yet not central information related to a search, goes missing if a chatbots is involved. The bot filters information that is not directly linked to the query. However, sometimes the knowledges that are found by chance are crucial to the search result as they provide important further information or enable the user to better pin down the knowledges and to understand them in their contextuality.

The term *sealed knowledges* refers to the singling out of one answer of the potential, complex and context-dependent

answer space of a search query. While it is the knowledge that is *sealed*, the metaphorical *seal* the term refers to is the LLM which functions like a very narrow channel. The large amount of possible information and knowledges is heavily filtered through the algorithm. The language model, by filtering what it outputs, is the specific point of the seal. Only little knowledge passes through the algorithm and is shown to the user in the form of a simplified, singular, and easy answer. This *seal* can and needs to be broken or circumvented to reach the complex knowledge space behind it. The answer the language model assembles is the visible result of the *sealing of knowledges*.

This is different to ‘traditional’ web search in which a (decisive) structuring, yet not a sealing of knowledges and the knowledge space takes place. While the algorithms in ‘traditional’ search, too, mitigate which websites are shown on which page of search results and with that which ones are most likely to be engaged with, several websites are still shown on the first page of a search results, and users actively decide on which one they click (and on which one not). They may be unsatisfied by one website and decide to click on another one, find information they were not explicitly looking for but that turns out to be essential for their search, or realize that a website is heavily biased and does not seem to be trustworthy. Moreover, when entering a website, users can acquire some knowledge about the platform, making it possible to get an idea of the context in which the information was written. In ‘traditional’ online search, the output is thus *structured* and *sorted*. However, it is not *sealed* in the same way that the output of a chatbot integrated into online search seals the possible knowledge space of an algorithmically produced answer. Additionally, as I will argue in Sect. 3.2., direct answers in online search create specific user subjectivities through the *sealing of knowledges* which is different than the user subjectivities produced by ‘traditional’ online search.

The term *sealed knowledges* is both a description and a critique of the current trend to increasingly integrate chatbots into search engines. As I will show in the following, the sealing of knowledges has a twofold impact through which it becomes a qualitatively new phenomenon from ‘traditional’ web search. First, it implicitly shifts the understanding of knowledges in online searches and second, it has an impact on user subjectivities.

3.1 (De)situated knowledges

The first implication of the *sealing of knowledges* through the usage of chatbots providing direct answers in online search is their reliance on—and reinforcement of—a view of knowledge as de-situated and disembodied. When chatbots are trusted to provide answers to search queries, it implies that they are expected to provide good, sufficient, and ‘true’

answers. It seems to mean that LLMs deliver knowledges just as well as websites in ‘traditional’ online search. Knowledge (in singular) and information in this understanding is factual, clear, and unambiguous. It can be taken out of its original context (which is what the algorithm does when scraping the internet for an answer to the search query) without losing any of its value. That includes a disregard of further knowledges, discourses, or controversies around a specific piece of information which may be part of its original embeddedness. Knowledge is therefore thought of as context independent by the people implementing a chatbot for the provision of direct answers.

When being portrayed with a direct answer to a search query, users read an information that is broken down into a discernible text. For example, Fig. 3 shows the answer to the question “what is racism?” in Bing Copilot.

The output of the algorithm appears highly structured, with highlighted headings (in bold) and a list which points to several different aspects concerning racism. Without going into the details of the content of the output, this algorithmically assembled text provides a certain view on what racism is. As an important, long-standing, and contested concept, the answer the LLM produces is necessarily only one of several possible answers that could be given to the search query “what is racism?”. This, arguably, is mostly the case when looking up information. If I would type the same search query into a ‘traditional’ search engine and visit a website on which a person has written an explanation of racism, that information would also be partial. What is new and different through the usage of chatbots is that the information does not result from a situated knowledge which is broken down to data/text that can followingly be transferred. As already explained in Sect. 2.2., information describes the text in itself and the content it carries, which lacks context and application. It is processed data which can be transferred. In traditional search, this information is assembled by a human who holds it as knowledge which is embodied in a situated subjectivity. It is a knowledge which is embedded in the lived experience of a person who can abstract it to write it down as transferrable information in a text. Through using chatbots in search engines, this changes. The text is assembled by a disembodied and de-situated algorithm (see below). The algorithm has never lived in a racist society, has never been subjected to racist slurs and has never engaged in a racist activity. An embodiment, situatedness and lived experience is exactly what the language model does not have. Thus, the use of chatbot for direct answers changes how information and the concurrent knowledge is created.

This de-situated assemblage of a search answer marks a stark contrast to the idea of *situated knowledges* by Haraway and the attention it draws to the partiality, embodiment and situatedness of knowledges outline above. If I am presented with an LLM generated answer to my search query, I do not

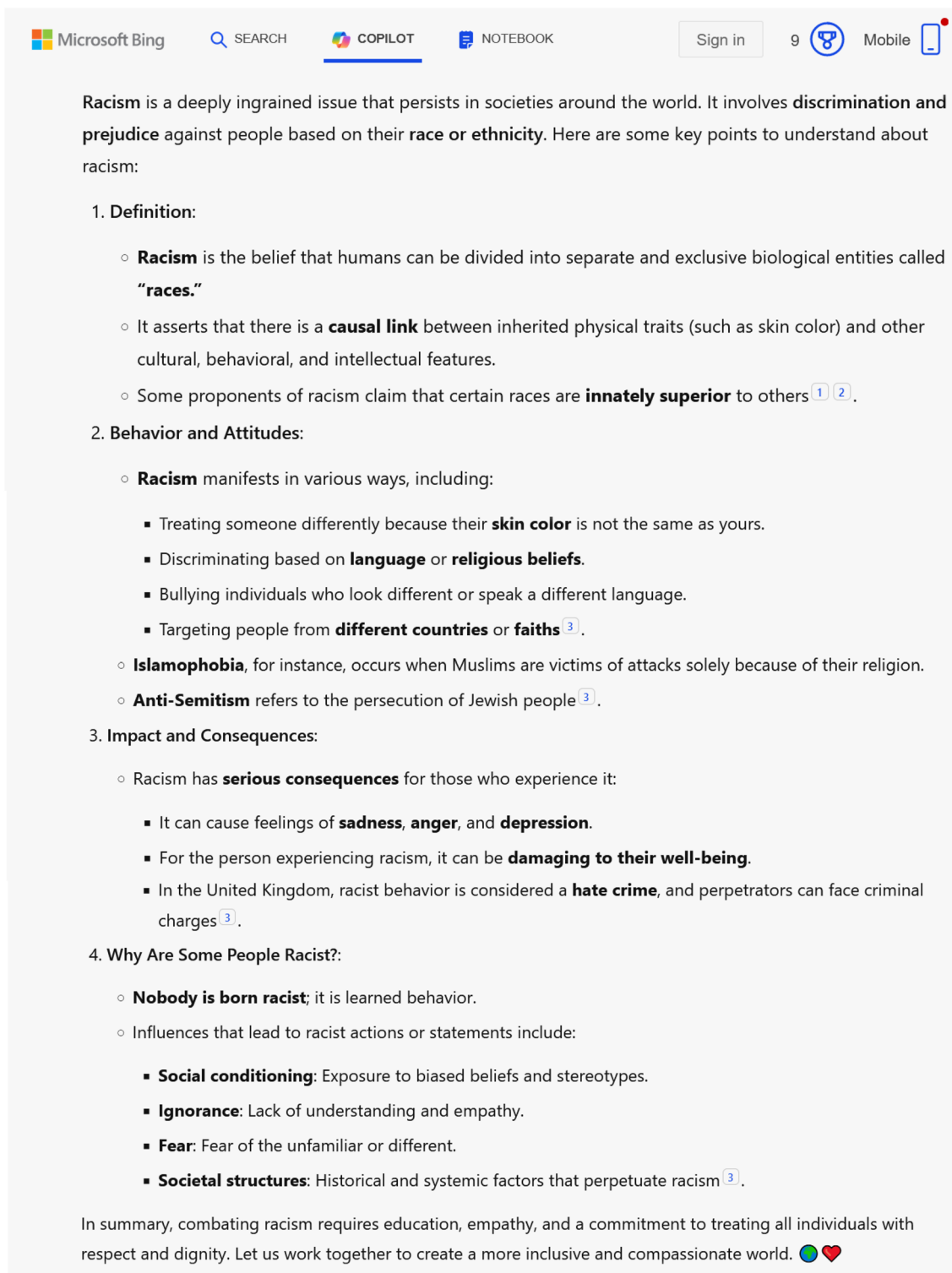


Fig. 3 Screenshot, March 2024. Bing’s chat interface when the question “What is racism?” is entered

know where the knowledge originally came from. The direct answer to a search query is an algorithmically produced text which—in most cases—assembles and re-arranges knowledge from several different sources. The original text,

written from a situated perspective, ceases to exist when an algorithm combines it with other texts to produce an output. It is difficult for users to know where the information which they are shown originates. Even if there is a link to one or

more websites from which the information originates and users can follow it to check the source, these links are not necessarily the only source of knowledge from which the direct answer is compiled. LLMs are trained on huge text corpora. Through this training, they simulate both a specific style of writing (namely the one which is dominant in the training data) and may also reproduce some of the content on which it was trained. The training data, in turn, often remains hidden as companies often do not make them public. This makes the text production and knowledge generation through the LLM even more opaque.

The de-situatedness of the algorithmically produced text, as a non-situatedness in a certain context, discourse, and original space of writing, is reinforced by the inherent disembodiment of the language model. As Bender and Koller (2020) write, LLMs miss any kind of bodily situatedness. If a person writes a text which is then published online, for example a Wikipedia entry, they have the capacity to reflect on their writings and the sources which they use to write the text. They also have an embodied experience of being in the world. Language models, on the other hand, while necessitating a material infrastructure of servers, etc., are disembodied and have no (embodied) understanding of the world. Therefore, they lack even the possibility of giving an embodied account of knowledges. Bender and Koller (2020) point out that while LLMs may be very good at simulating the *form* of human language, they do not have any way of ‘learning’ *meaning* from it (Bender and Koller 2020). This is due to their technological functioning: they are programmed to output statistically relevant and likely words and texts. They do this by imitating patterns of words and texts with which they are trained. The algorithms imitate the form of the human text without any possibility for understanding meaning it. Thus, using chatbots for the provision of direct answers in online search presupposes that knowledge is conceived of as disembodied, as it is assembled by a disembodied AI system.

The usage of LLMs to generate direct answers therefore presupposes a conception of knowledge as disembodied, impartial, and non-situated or de-situable which stands in stark contrast to feminist epistemologies and to theories of *situated knowledges* by Haraway. As users of the bot get used to retrieving information with it, they get more and more accustomed to this conception of knowledges.

3.2 User subjectivities

Secondly, the implementation of chatbots into search engines relies on a very specific, imagined user interacting with it. This, through the following *sealing of knowledges*, leads to user subjectivities which adhere to the previously imagined user. First to the imagined user. Metzler et al. (2021), a team of Google researchers, wrote an opinion paper about the

future of the Google search in 2021. In it, they lay out the idea to use LLMs for the generation of direct answers and proclaim that this mode of search will be the way forward, foreshadowing the current development of online search. The reasons they name for why chatbots should be implemented in online search are telling about the image they have on the users engaging with the search engine. They proclaim that ‘traditional’ web search “induces a rather significant cognitive burden on the user” which would be resolved if chatbots provided one simple, clear and authoritative answer to user inputs (Metzler et al. 2021, p. 1). The formulation of searches placing a “cognitive burden” on users is quite remarkable. It implies that users cannot, and do not want to, choose between different websites. Even more, choosing between different answers is framed as negatively impacting users and as potentially even being harmful for them (a *burden* is quite negatively connotated term). This negative aspect of online search, so Metzler et al., can be alleviated through the integration of chatbots into search engines. In contrast to ‘traditional’ search, the integration of chatbots into search engines is framed as “user-friendly”, “convenient” and “time-saving” (Shah and Bender 2022, p. 222). The authors’ idea of what users want in this proposal (which has now become reality) is, therefore, to receive a simple and clear answer in order to save time.

This imagination of users has several implications for the implementation of chatbots into online search. First, the necessity of providing users with the information of how the LLM works, or of the *sealing of knowledges* taking place through it, is not a matter of debate. Even more, it is not mentioned (Metzler et al. 2021; Pichai 2023). In the Bing chat function, there is no attempt to clarify to users why they are shown a certain output or how the algorithm works. Due to the imperative of saving time and maximal convenience, users are not expected to question results they receive or the way an answer to their search query is assembled by the language model. The user is not expected to want to understand how the system functions—very much in line with the theory of *sealed surfaces* by Mühlhoff. This is closely connected to the specific interface design of the search engines. Figure 4 shows the start page of the Bing chat feature. The sleek interface does not invite a critical engagement with the system but rather seems to spread an aura of trustworthiness. There is no tab telling the user how the system works or cautioning against taking the output of the bot as the only possible answer (which it is often not, as we saw in Figs. 1 and 2). The chat interface reads “ask me anything” (Fig. 4). The chatbot is anthropomorphised (“me”) and at the same time has an appearance of omniscience (“anything”). It promises to provide better answers than ‘traditional’ search. The whole appearance, both content-wise and design wise does not tell how the bot functions and how answers are produced. The *sealed surface* (Mühlhoff Forthcoming) of

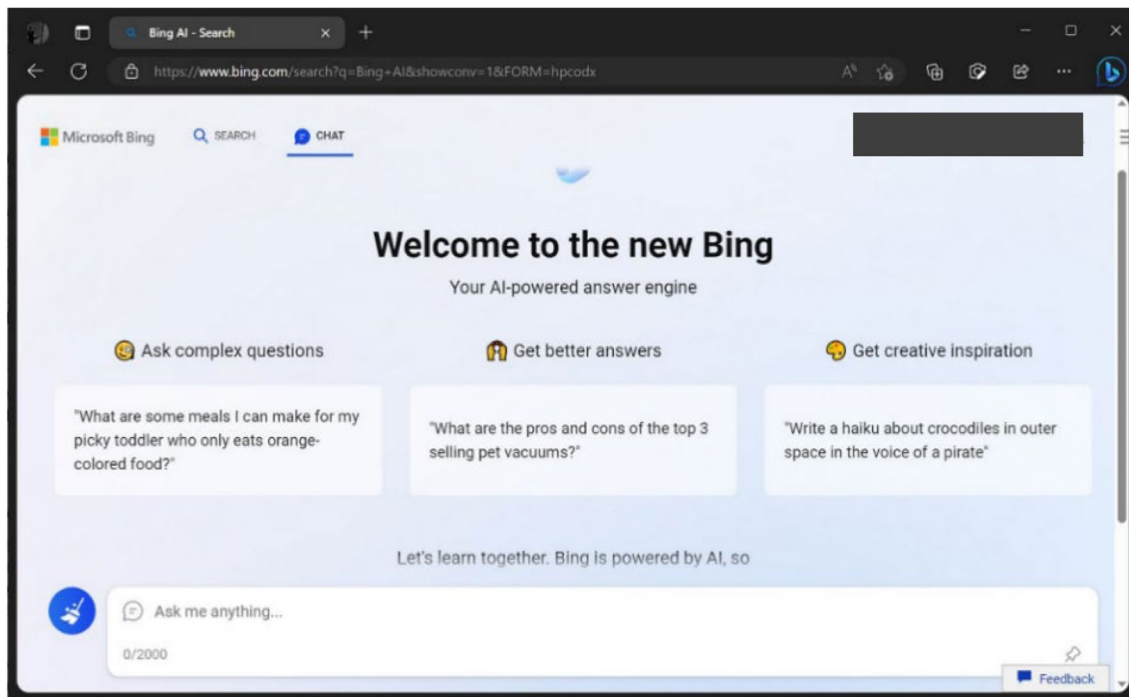


Fig. 4 Screenshot, September 2023. Bing chat interface

the interface design of the LLM integrated into the online search engine feeds into the *sealing of knowledges*.

Secondly, this imagined idea of users and their wishes, and the concurrent implementation of the search based on this as described above, impacts users and shapes their subjectivities. I understand subjects and subjectivation based on theories by Foucault here. Very briefly put, *subjects*, in this theoretical understanding, are constituted by power relations and emerge in and through them (see Kammler et al. 2014, p. 294). Subjects are in constant *subjectivation processes* through which they are constituted and produced. Subjectivation processes can be understood as the interplay between the power relations that produce the subject and the formation of the subject's own power relations (Vogelmann 2017, pp. 11–12). The integration of chatbots into search plays into the subjectivation processes of the users engaging with them. It creates a user subjectivity in which the display of a singular answer to a search query is taken as the norm and passed unquestioned. It becomes increasingly difficult to question how the system works as it becomes the norm to accept the answer of a bot without a question. It is convenient and fast to use the bot and there is nothing which shows the potentially problematic implication of taking its answer as authoritative. The user subjectivity is centered around mere acceptance of, not critical engagement with, the answers of the bot. This, in a larger realm, can lead to subjects which increasingly accept singular answers instead of questioning the information they find to build their own

knowledge. The *imagined* user, through the actual implementation of this new mode of search, shapes the subjectivation and subjectivities of *actual* users and thus creates lived realities.

The implementation of chatbots in searches increasingly takes away users' control over what information they can find in a search. Users do not have the possibility to click on different search results and concurrently do not find different opinions on an issue by chance. While the search results may state several different points concerning an issue, it is still a less diverse picture than in 'traditional' search, where different opinions are voiced quite strongly on different websites by different people. It may, at times, be unpleasant for users to engage in certain discourses by being thrown into these types of discussions, but it is important to get an overview over complex and ambiguous issues. Having to decide which websites to visit and having to put effort into finding certain information is pivotal to critically build an own opinion and to form informed personal knowledges.

3.2.1 Relevance

This creation of user subjectivities through implementation and design choices carries further relevance. The direct answers given by a chatbot not only seal knowledges and are based on a de-situated understating of knowledges, but they are also prone to include misinformation and to reproduce hegemonial knowledges. If subjects receiving direct answers

increasingly do not question the output of the chatbots, as part of their emerging subjectivity, this is problematic. In 2023, AlgorithmWatch and AI Forensics, two organizations investigating the impact AI algorithms, did a joined study in which they analysed the answers Bing chat gave to questions regarding the state elections in Bavaria, Hesse, and Switzerland (AlgorithmWatch and AI Forensics, October 5th, 2023). They found that the answers the chatbot gave, even to presumable ‘easy’ question like “who are the top candidates of each party in the election in Hesse 2023?”, were plainly wrong. This is dangerous, and vividly shows the political implication the integration of the chatbot can have, as it spreads misinformation and may heavily impact on the voting behaviour of people trying to access information through the bot. Knowledges about the elections which are accessible via “traditional” web searches, were filtered through the chatbot algorithm and not displayed. They have been sealed by the algorithm providing a direct answer. A seal that is increasingly difficult to break through the creation of user subjectivity which does not include a questioning and critical engagement with the answer of the algorithm.

Even when the answer of such a bot does not spread clear misinformation, it is inherently prone to reproduce hegemonic knowledges and seal marginalised knowledges. It is often the less-dominant views, the marginalized views, the ones spoken from the margins, the ones spoken in resistance, which are not heard in the dominant discourses (see hooks 1989). This is reproduced in chatbots giving direct answers to search queries. Through the increasing *sealing of knowledges*, marginalized voices and views may be even more difficult to find. An example for this are the search results in Fig. 2, in which a dominant idea of who famous philosophers are is replicated. Hegemonic voices dominantly structuring the discourse and are most likely to be found in online search. They are voices which can be found e.g. on Wikipedia, which make up a large part of the content of most LLM training data and are likely to be included in the answers of chatbots as search engines (see Bender et al. 2021).⁴ While AI algorithms are often portrayed as neutral and unbiased, they reflect the socio-cultural biases which are imbedded in

their design and in the data on which they are trained. Algorithms are deeply entrenched in the historical, cultural and social situation out of which they emerge (Amershi 2020). Chatbots assembling direct answers to search queries are no exception to this, they also reflect social and cultural biases and dominant narratives in their output. While the output a chatbot provides may seem self-explanatory, knowledge and “information is not self-explanatory; it is context dependent. To be useful—or at least meaningful—it must be understood through the lenses of culture and history” (Kissinger et al. 2021, p. 52).

The increasing *sealing of knowledges*, through the usage of chatbots providing direct answers to search queries, therefore leads to a solidification of hegemonic knowledges and even to blatant misinformation. This can have serious political and societal consequences. As more and more people turn to chatbots assembling direct answers to search queries, misinformation can, therefore, spread even faster than before in society. Simultaneously, through the repetition of hegemonic knowledges, imagining different ways of living and the experiences of marginalized people may become more and more difficult. This may impact on voting behaviour (as shown above) and solidify existing societal power structures and knowledges.

4 Outlook: Re/introducing possibilities of doubt

4.1 Algorithms and the reduction of doubt

The provision of a direct answer by chatbots and the co-occurring *sealing of knowledges* leads to a reduction of possibilities of doubt. At the same time, doubt can be a crucial way of resisting the *sealing of knowledges*. In her book “Cloud Ethics”, the political geographer Louise Amoore (2020) describes how algorithms change what comes into perception. She argues that algorithms reduce and distill certain outputs from vast amounts of data, thus bringing certain, actionable outputs to the forefront of attention. Other possible actions which could take place are disregarded. Algorithms thereby control which structures in data become visible—and which ones not. They determine for example who is flagged as a possible suspect at a border control. In the moment of flagging, the singled-out person is brought to the attention and into conscious perception of the border control. Therefore, algorithms promise to reduce uncertainty by providing an actionable output in situations of uncertainty about what to do and how to act. They reduce the doubt which otherwise accompanies decisions. The opaque algorithms seem to produce clarity by giving a singular answer to complex questions based on vast amounts of data.

⁴ In January 2023 alone, Wikipedia had more than 23 billion page views (Wikimedia Statistics). It is a main point of reference for knowledge search online and contains hundreds of thousands of articles in various languages and is continuously growing its article numbers. It is the biggest online encyclopaedia—and written from a writer base consisting mainly of white men from an educated background. Because of its communal structure, its writer base has the power to normalize a specific way of writing on the platform which signals a certain political outlook (e.g. through forbidding the usage of gendered language). However, Wikipedia is often a main part of the knowledge-base for LLMs, which is then likely to reproduce both the writing style and the contents of their training data (Bender et al. 2021), thus reproducing its knowledges and discourses.

While Amoores mainly refers to automated border control and surveillance algorithms, some of her claims can also be applied to automated text-generation in the usage of chatbots giving direct answers to search queries. Through LLMs, certain knowledges are brought to attention of users, while other possible knowledges out of the complex knowledge space are disregarded. The algorithm determines what becomes perceptible, findable knowledges and opens spaces to act on these, and only these, algorithmically filtered knowledges. Through the automatically assembled output, the underlying information selection mechanisms and text-generation processes are opaque and invisible. This reduces the room for doubt, as doubting the algorithms presupposes and necessitates the knowledge that doubting the algorithm is *possible*. This, as I wrote in Sect. 3.2., becomes increasingly difficult through the creation of specific user subjectivities which do not include a desire for understanding how the LLM works and is aimed at an unquestioned acceptance of the output the bot produces.

4.2 Doubting the output of algorithms

However, even though chatbots as search engines may reduce doubt through the creation of specific user subjectivities, they do not eliminate the possibility of it. In reference to Amoores, I suggest that we need to start and keep doubting the algorithm and the output of automated answers to search inquiries. Amoores (2020, p. 24) argues that.

cutting against the grain of the dominance of definiteness as algorithms act on doubt, I seek to reinstate doubtfulness [...] within the calculative architecture of the algorithm. Though at the point of optimized output, the algorithm places action beyond doubt, there are multiple branching points, weights, and parameters in the arrangements of decision trees and random forest algorithms, branching points at which doubt flourishes and proliferates.

This doubtfulness works in manifold ways. It is a doubt of oneself and how oneself knows, as well as a doubt of how the algorithm works and why it displays certain answers. It is a doubt of the singularity of the answer the algorithm provides and of the idea of a securable future which it promises (Amoores 2020, p. 148). It is a doubting of data points and a doubtful accounting of the calculations of the algorithm as well as a doubting of the perspectives which are shown through the algorithms and of their inherent partiality.

In the specific context of chatbots providing direct answers to search queries, doubting means to question the validity and adequacy of the output of the search answer as being the one and only correct—or at least most relevant—one. Even though the *sealing of knowledges* makes it more difficult to access the diverse possible answer space to a

search query, it is still there. The sleek, intuitive user interfaces may seal knowledges from view. But it does not eliminate those knowledges. It just makes it harder to find them. A seal can be broken. Doubting requires looking for cracks in the seal, to widen them, to build a sense for them. It means the will to *unseal* knowledges, to dig behind shiny surfaces, to look for the knowledges flourishing behind it. It involves doubting the generalization of the portrayed knowledges and an awareness for the inherent situatedness of the portrayed knowledges which were taken from somewhere and are now newly arranged, de-situated, nowhere in particular. It means to be aware that knowledges are always partial and stand in certain contexts—and that the output of a LLM is therefore also always partial and context dependent. To refer back to Haraway (1988): acknowledging and cherishing the situatedness and partiality of knowledges is what we need to do to avoid the *god trick*. Doubting thus involves a conscious re-situating of knowledges.

Situating knowledges is a strategy to become aware of the dominance of certain knowledges and views of knowledge over others. It allows to see contingencies in portrayed knowledges. It means, to draw on the famous quote by Haraway (2016), to stay with the trouble, to choose the way of leaving the future open by being aware of the multiplicity of knowledges in the present instead of “making an imagined future safe” (Haraway 2016, p. 1). It means to take the troublesome way of doubting the knowledges and outputs portrayed by chatbots and to question what it sealed. This means to reinstall an understanding of knowledges in their situatedness and partiality, to imbed knowledges in discourses and power-relational emergences. Doubting the outputs of LLMs can function as a feminist intervention to resist the sealing of knowledges, especially of marginalized and non-dominant knowledges (see e.g. Ahmed 2008; Haraway 2016). Writing about feminist futures, Sara Ahmed (2008, p. 251) stresses the importance of hope, “a hope that things can be different, and that the world can take different forms”. This idea of open, non-determined futures, allowing for hope, different imaginaries, and political action in the present, crucially includes a future that is not pre-determined by algorithmically given answers which filter which knowledges come to be known.

5 Conclusion

In this paper, I focussed on the impact of the usage of chatbots providing direct answers to search queries in online searches on knowledges and epistemic possibilities in society. I argue that this new mode of online search leads to a *sealing of knowledges*. The *sealing of knowledges* describes the implicit breaking down and simplifying of a complex answer space to only one answer. This is based

on an imagination of users who want to get to knowledge as conveniently as possible and do not want to put ‘effort’ into looking at several answers to a search query. The implementation of chatbots providing direct answers based on this *imagined* user creates user subjectivities of *actual* users who are increasingly prone to uncritically accept a singular, convincing answer as true, “objective” and conclusive, even though it is partial, situated and only one possible way to answer the search query. This is problematic, as algorithmically assembled answers are likely to spread misinformation, reinforce hegemonial knowledges, and often do not include marginalized knowledges.

Moreover, the provision of direct answers and the *sealing of knowledges* implicitly relies on a de-situated, disembodied, context independent and impartial understanding of knowledges. In the usage of chatbots providing direct answers, chatbots take knowledges out of context and re-assemble them. Through this decontextualization, discourses surrounding a topic are obscured and knowledges dislocated. The *sealing of knowledges* and the corresponding user subjectivity may lead to a reduction of doubt of the search answer. However, despite this seeming reduction of doubt, it is possible to doubt the output of an algorithm and to question which knowledges are sealed through their implementation. Doubt can be a productive force to re-situate knowledges, to understand search results in their contingency, and to re-surface marginalized knowledges. By introducing the concept of *sealed knowledges*, I add a new way to conceptualize the impact of the usage of chatbots in search engines both on individuals and on society. This is so far missing in the academic debate. Furthermore, by discussing theoretical accounts of possibilities of doubting the outputs of algorithms, I show ways to resist the increasing *sealing of knowledges*.

We need to start engaging in a critical discourse on the *sealing of knowledges* through chatbots providing direct answers in search engines. The decision to include chatbots to produce direct answers into search engines has a profound impact on knowledges and possibilities of knowing in society, and it is important to start an informed public debate about it. Search engines are a central part of our knowledge infrastructure. They decisively shape which knowledges can be found and accessed, and how subjects develop and cultivate their epistemic capacities. With that, they have serious political and societal consequences, and occupy a position of power within society. Therefore, the questions of how they are designed and who decides how they are implemented has far reaching consequences. While the proposed concept of *Sealed Knowledges* helps to understand the implications of this new mode of online search, it poses urgent further philosophical and political questions. It is time to think about which power search engine companies yield over knowledge infrastructures, whether this situation is desirable, and

whether it is time to implement regulatory measures which limit the power of search engine companies.

Moreover, we need a growing awareness for the importance of media literacy, including knowledge on “the economic, political and socio-cultural dimensions of search engines” (van Dijck 2010, p. 574). Using chatbots as search engines promises to make information easier available, to provide more useful information in a shorter amount of time and to lower barriers to access of knowledges. However, not everything that is shiny is gold, and just because using chatbots to produce direct answers in search engines sounds like a promise of convenience, we should be cautious whether we want to buy into the hidden narratives allowing for this technology.

Acknowledgements I would like to thank the members of the “Ethics and Critical Theories of AI” colloquium, especially Paul Schütze, Jan Siebold, Imke von Maur and Rainer Mühlhoff, for their helpful comments on earlier drafts of this paper.

Funding Open Access funding enabled and organized by Projekt DEAL. No funding was received to assist with the preparation of this manuscript.

Data availability No data has been used for this article.

Declarations

Conflict of interest The author has no relevant financial or non-financial interests to disclose.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article’s Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article’s Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Ahmed S (2008) Feminist futures. In: Eagleton M (ed) Blackwell concise companions to literature and culture. A concise companion to feminist theory. Blackwell Publishing, p 236–254
- AlgorithmWatch and AI Forensics (2023) ChatGPT and Co: are AI-driven search engines a threat to democratic elections? October 5. <https://algorithmwatch.org/en/bing-chat-election-2023/>
- Amershi B (2020) Culture, the process of knowledge, perception of the world and emergence of AI. AI & Soc 35(2):417–430. <https://doi.org/10.1007/s00146-019-00885-z>
- Amoore L (2020) Cloud ethics: algorithms and the attributes of ourselves and others. Duke University Press

- Anderson E (2020) Feminist epistemology and philosophy of science. The Stanford Encyclopedia of Philosophy. <https://plato.stanford.edu/entries/feminism-epistemology/#toc>
- Bender EM, Koller A (2020) Climbing towards NLU: on meaning, form, and understanding in the age of data. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, p 5185–5198.
- Bender EM, Gebru T, McMillan-Major A, Mitchell M (2021) On the dangers of stochastic parrots: can language models be too big? In: Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency. <https://doi.org/10.1145/3442188.3445922>
- Google. Gemini: supercharge your creativity and productivity. <https://gemini.google.com/>
- Gusbeth S (2023) Baidu entwickelt einen Konkurrenten für ChatGPT. Handelsblatt, January 30. <https://www.handelsblatt.com/technik/it-internet/kuenstliche-intelligenz-baidu-entwickelt-einen-konkurrenten-fuer-chatgpt/28951800.html>
- Haraway D (1988) Situated knowledges: the science question in feminism and the privilege of partial perspective. *Fem Stud* 14(3):575–599. <https://doi.org/10.4324/9781003001201-36>
- Haraway D (2016) Staying with the trouble: making Kin in the Chthulucene. Duke University Press.
- Harding S (1986) The science question in feminism. Cornell University Press
- hooks b (1989) Choosing the margin as a space of radical openness. *Framework: J Cine Media* 36:15–23.
- Huang Z (2023) Chinese search Giant Baidu to Launch ChatGPT-style bot. Bloomberg, January 30. <https://www.bloomberg.com/news/articles/2023-01-30/chinese-search-giant-baidu-to-launch-chatgpt-style-bot-in-march?leadSource=verify%20wall>
- Introna LD, Nissenbaum H (2000) Shaping the web: why the politics of search engines matters. *Inf Soc* 16:157–173. <https://doi.org/10.4324/9781315259697-19>
- Kammler C, Parr R, Schneider UJ, Reinhardt-Becker E (eds) (2014) Foucault Handbuch: Leben—Werk—Wirkung. J.B. Metzler. <https://doi.org/10.1007/978-3-476-01378-1>
- Kissinger HA, Schmidt E, Huttenlocher D (2021) The age of A.I.: and our human future. John Murray (Publishers)
- Lindern J von (2023) Microsoft baut Chatbot in seine Suchmaschine ein. *Zeit Online*, February 7. <https://www.zeit.de/digital/2023-02/microsoft-bing-chatgpt-ki-suchmaschine/seite-2>
- McLaren MA (2002) Feminism, Foucault, and embodied subjectivity. SUNY series in contemporary continental philosophy. State University of New York Press
- Metzler D, Tay Y, Bahri D, Najork M (2021) Rethinking search: making domain experts out of dilettantes. *ACM SIGIR Forum* 55(1):1–27. <https://doi.org/10.1145/3476415.3476428>
- Milmo D (2023) Microsoft confirms multibillion dollar investment in firm behind ChatGPT. The Guardian, January 23. <https://www.theguardian.com/technology/2023/jan/23/microsoft-confirms-multibillion-dollar-investment-in-firm-behind-chatgpt>
- Mühlhoff R (2018) Digitale Entmündigung und user experience design: Wie digitale Geräte uns nudgen, tracken und zur Unwissenheit erziehen. *Leviathan—Berliner Zeitschrift Für Sozialwissenschaft* 46(4):551–574. <https://doi.org/10.5771/0340-0425-2018-4-551>
- Mühlhoff R (Forthcoming) Ethics of AI in the digital media age: power, critique, responsibility
- Nayak P (2019) Understanding searches better than ever before. Google. <https://blog.google/products/search/search-language-understanding-bert/>
- Pichai S, Hassabis D (2023) Introducing Gemini: our largest and most capable AI model, December 6. <https://blog.google/technology/ai/google-gemini-ai/#sundar-note>
- Pichai S (2023) An important next step on our AI journey. Google, February 6. <https://blog.google/technology/ai/bard-google-ai-search-updates/>
- Potthast M, Hagen M, Stein B (2020) The dilemma of the direct answer. *ACM SIGIR Forum* 54(1):1–12. <https://doi.org/10.1145/3451964.3451978>
- Reputation911 (2023) Do people click on the second page of Google? February 23. <https://reputation911.com/do-people-click-past-the-first-page-of-google/>
- Röhle T (2010) Der Google Komplex: Über Macht im Zeitalter des Internets. transcript.
- Segev E (2008) Search engines and power: a politics of online (mis-) information. *Webology* 5(2):1–18
- Shah C, Bender EM (2022) Situating search. *CHIIR '22*, March 14–18, p 221–232. <https://doi.org/10.1145/3498366.3505816>
- Southern MG (2020) Over 25% of people click the first google search result. <https://www.searchenginejournal.com/google-first-page-clicks/374516/>
- Stanley L, Wise S (2013) Method, methodology and epistemology in feminist research processes. In: Stanley L (ed) *Routledge library editions: feminist theory: vol. 13. feminist praxis: research, theory and epistemology in feminist sociology*. Routledge, p 20–60
- Stockdale K (2023) Thanks for helping me make a point about philosophy and gender, Chat GPT [Tweet]. Twitter, February 27. <https://twitter.com/katiestockdale/status/1630241137721221121>
- Traweek S (1988) Beamtimes and lifetimes: an ethnography of the high energy physics community in Japan and the US. Harvard University Press.
- van Dijck J (2010) Search engines and the production of academic knowledge. *Int J Cult Stud* 13(6):574–592. <https://doi.org/10.1177/1367877910376582>
- Vogelmann F (2017) Foucault lesen. Springer Fachmedien Wiesbaden. <https://doi.org/10.1007/978-3-658-15474-5>
- Waldby C (1995) Feminism and method. In: Pringle R (ed) *Transitions: new Australian feminisms*. Routledge
- Wikimedia Statistics. Total page views. <https://stats.wikimedia.org/#/all-projects/reading/total-page-views/normal%7Cbar%7C2-year%7C~total%7Cmonthly>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.